



Universidad Modelo
Escuela de Ingeniería

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Ingeniería en Desarrollo de Tecnología y Software

Administración de Bases de Datos

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Práctica 2

1. Creación de la imagen y el contenedor de Docker.

```
Practica2_PetCareSolutions — zsh — 166x36

[(base) millie@Millies-MacBook-Pro Practica2_PetCareSolutions % docker build -t petcare_image .
[+] Building 1.1s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 410B
=> [internal] load metadata for docker.io/library/mysql:8.0
=> [auth] library/mysql:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/mysql:8.0@sha256:7dcdde01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
=> => resolve docker.io/library/mysql:8.0@sha256:7dcdde01f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
=> [internal] load build context
=> => transferring context: 1.39kB
=> CACHED [2/5] COPY 1_creacion_db.sql /docker-entrypoint-initdb.d/
=> [3/5] COPY 2_creacion_tablas.sql /docker-entrypoint-initdb.d/
=> [4/5] COPY 3_creacion_relaciones.sql /docker-entrypoint-initdb.d/
=> [5/5] COPY 4_registros.sql /docker-entrypoint-initdb.d/
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:0a089c8a2f7d2b08218c7bbb7f966b807dcb1d1d29ba54f48333783444c4da72
=> => exporting config sha256:ef470e15eca676de8abf676bf1d071498394efe055e4f95a6cbf26cb59d40d6
=> => exporting attestation manifest sha256:4a219fbf2fa469d0be3f136445794fd2b96a2679cc7cb1d0374082171993a65
=> => exporting manifest list sha256:a26be140f3fbf758f1c06e781560a72260b47287d7beca14ca4af2f7d3c5a012a
=> => naming to docker.io/library/petcare_image:latest
=> => unpacking to docker.io/library/petcare_image:latest

1 warning found (use docker --debug to expand):
- SecretsUsedInArgOrEnv: Do not use ARG or ENV instructions for sensitive data (ENV "MYSQL_ROOT_PASSWORD") (line 3)

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/a6a7h9t3un7vbihk55xklhym
[(base) millie@Millies-MacBook-Pro Practica2_PetCareSolutions % docker run -d --name petcare_container -p 3308:3306 petcare_image
4751d66f839475d7fc1b073d1221300c385b57b5e0e861476de2a4aa2a747917
(base) millie@Millies-MacBook-Pro Practica2_PetCareSolutions % ]
```

2. Acceso al contenedor y al entorno de MySQL.

```
Practica2_PetCareSolutions — docker exec -it 4751d66f8394 bash — 166x36

[(base) millie@Millies-MacBook-Pro Practica2_PetCareSolutions % docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED      STATUS      PORTS                               NAMES
4751d66f8394   petcare_image   "docker-entrypoint.s..." About a minute ago   Up About a minute   0.0.0.0:3308->3306/tcp, [::]:3308->3306/tcp   petcare_container
[(base) millie@Millies-MacBook-Pro Practica2_PetCareSolutions % docker exec -it 4751d66f8394 bash
[bash-5.1# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> ]
```

3. Verificación de la creación de la base de datos y sus respectivas tablas.

```
Practica2_PetCareSolutions — docker exec -it 4751d66f8394 bash — 166x36

[mysql> show databases;
+-----+
| Database |
+-----+
| garcia_almaguer |
| information_schema |
| mydatabase |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.01 sec)

[mysql> use garcia_almaguer;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
[mysql> show tables
+-----+
| Tables_in_garcia_almaguer |
+-----+
| atenciones_medicas |
| clientes |
| mascotas |
| medicamentos |
| prescripciones |
| veterinarios |
+-----+
6 rows in set (0.00 sec)

mysql> ]
```

4. Verificación de las relaciones establecidas entre las tablas

```
Practica2_PetCareSolutions — docker exec -it 4751d66f8394 bash — 166x36

mysql> select
  -> table_name,
  -> column_name,
  -> constraint_name,
  -> referenced_table_name,
  -> referenced_column_name
  -> from information_schema.key_column_usage
  -> where table_schema = 'garcia_almaguer'
  -> and referenced_table_name is not null;
+-----+-----+-----+-----+-----+
| TABLE_NAME | COLUMN_NAME | CONSTRAINT_NAME | REFERENCED_TABLE_NAME | REFERENCED_COLUMN_NAME |
+-----+-----+-----+-----+-----+
| atenciones_medicas | id_mascota | fk_atenciones_mascotas | mascotas | id_mascota |
| atenciones_medicas | dni_veterinario | fk_atenciones_veterinarios | veterinarios | dni_veterinario |
| mascotas | dni_cliente | fk_mascotas_clientes | clientes | dni_cliente |
| prescripciones | id_atencion | fk_prescripciones_atenciones | atenciones_medicas | id_atencion |
| prescripciones | codigo_medimento | fk_prescripciones_medicamentos | medicamentos | codigo_medimento |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> ]
```

5. Verificación del registro de la información en las tablas.

```
Practica2_PetCareSolutions — docker exec -it 4751d66f8394 bash — 166x36

mysql> select * from atenciones_medicas;
+-----+-----+-----+-----+-----+-----+
| id_atencion | fecha_hora | diagnostico | costo_base | id_mascota | dni_veterinario |
+-----+-----+-----+-----+-----+-----+
| 1 | 2026-08-01 09:00:00 | Extraccion dental | 900.00 | 1 | 11111111A |
| 2 | 2026-08-02 10:30:00 | Cirugia de esterilizacion | 2000.00 | 2 | 33333333C |
| 3 | 2026-08-03 11:00:00 | Extraccion dental | 900.00 | 3 | 44444444D |
| 4 | 2026-08-04 14:00:00 | Consulta de rutina | 500.00 | 4 | 55555555E |
| 5 | 2026-08-05 16:00:00 | Cirugia de esterilizacion | 2000.00 | 5 | 22222222B |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql> select * from clientes;
+-----+-----+-----+-----+
| dni_cliente | nombre_completo | telefono | correo |
+-----+-----+-----+-----+
| 12345678A | Elsa Pato | 52451234 | elsa.pato@email.com |
| 23456789B | Susana Oria | 72555678 | susana.oria@email.com |
| 34567890C | Alan Brito | 59012 | alan.brito@email.com |
| 45678901D | Elvis Tek | 133456 | elvis.tek@email.com |
| 56789012E | Aquiles Castro | 5557890 | aquiles.castro@email.com |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from mascotas;
+-----+-----+-----+-----+-----+-----+
| id_mascota | nombre | especie | raza | fecha_nacimiento | dni_cliente |
+-----+-----+-----+-----+-----+-----+
| 1 | Shiro | Gato | Bombay | 2020-05-10 | 12345678A |
| 2 | Akira | Gato | Calico | 2019-03-15 | 23456789B |
| 3 | Hector | Perro | Pug | 2021-07-20 | 34567890C |
| 4 | Sofia | Perro | Pug | 2018-11-05 | 45678901D |
| 5 | Tomasa | Perro | Pug | 2022-01-30 | 56789012E |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
Practica2_PetCareSolutions — docker exec -it 4751d66f8394 bash — 166x36

mysql> select * from veterinarios;
+-----+-----+-----+-----+
| dni_veterinario | nombre_completo | especialidad | telefono |
+-----+-----+-----+-----+
| 11111111A | Dr. Armando Bronca | Veterinaria General | 912345678 |
| 22222222B | Dra. Elsa Capunta | Cirugia | 923456789 |
| 33333333C | Dr. Mario Neta | Veterinaria General | 934567890 |
| 44444444D | Dra. Marcia Ana | Veterinaria General | 945678901 |
| 55555555E | Dr. Alejandro King | Cirugia | 956789012 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from medicamentos;
+-----+-----+-----+-----+
| codigo_medimento | nombre_comercial | laboratorio | precio_unitario |
+-----+-----+-----+-----+
| 101 | Paracetamol | Laboratorio Guadalajara | 500.00 |
| 102 | Ibuprofeno | Laboratorio Itzamna | 800.00 |
| 103 | Alka Seltzer | Laboratorio Molina | 600.00 |
| 104 | Loratadina | Laboratorio A | 450.00 |
| 105 | Hidroxicina | Laboratorio Peru | 700.00 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from prescripciones;
+-----+-----+-----+-----+-----+
| id_prescripcion | id_atencion | codigo_medimento | cantidad | indicaciones |
+-----+-----+-----+-----+-----+
| 1 | 1 | 101 | 10 | Tomar 1 pastilla cada 8 horas por 5 dias |
| 2 | 2 | 102 | 5 | Tomar 1 pastilla cada 12 horas por 3 dias |
| 3 | 3 | 104 | 8 | Tomar 1 pastilla cada 12 horas por 4 dias |
| 4 | 4 | 105 | 6 | Tomar 1 pastilla cada dia por 6 dias |
| 5 | 5 | 103 | 1 | Tomar 1 pastilla cada 12 horas por 3 dias |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```